

Date- 03-12-2025

Data Normalization:

It is a process of organising the data in a database/data model to reduce redundancy and improve data integrity.

Advantages:

1. Remove duplicate data (Repeated data).
2. Improve data consistency
3. Make data easier to maintain
4. Reduce chance of update errors (upcoming errors)
5. improve storage efficiency

Data Modelling:

The process of connecting different data sources, cleaning and transforming data and building relationships between tables to create a structure and efficient data model for reporting and analysis.

In normalization the table will divided into two categories

- 1. Fact table:** This table will have measurable(numerical) or quantitative data.
- 2. Dimension table:** This table will contain unique categories data.

Cardinality:

Important: (Most dangerous cardinality is many to many) To solve - create one more table called as intermediate table (bridge table), it consists of unique department id.

Schemas:

Arranging all the tables in the proper order and connecting to the other table.

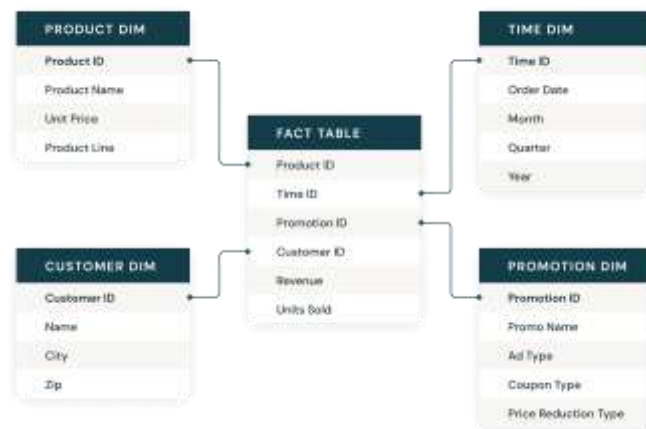
1. Star schema
2. Snowflake schema
3. Galaxy schema

Star schema:

- A star schema is a widely used data modelling technique in data warehousing that organizes data into a central fact table surrounded by multiple dimension tables.
- Its name derives from the star-like appearance of its structure in entity-relationship diagrams.

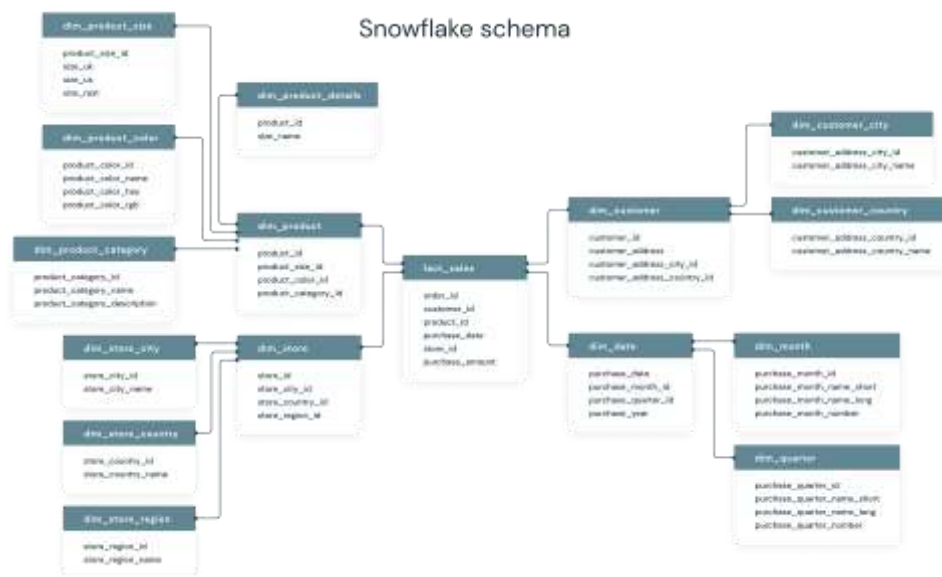
- This schema is designed to simplify data analysis and improve query performance.

Star schema



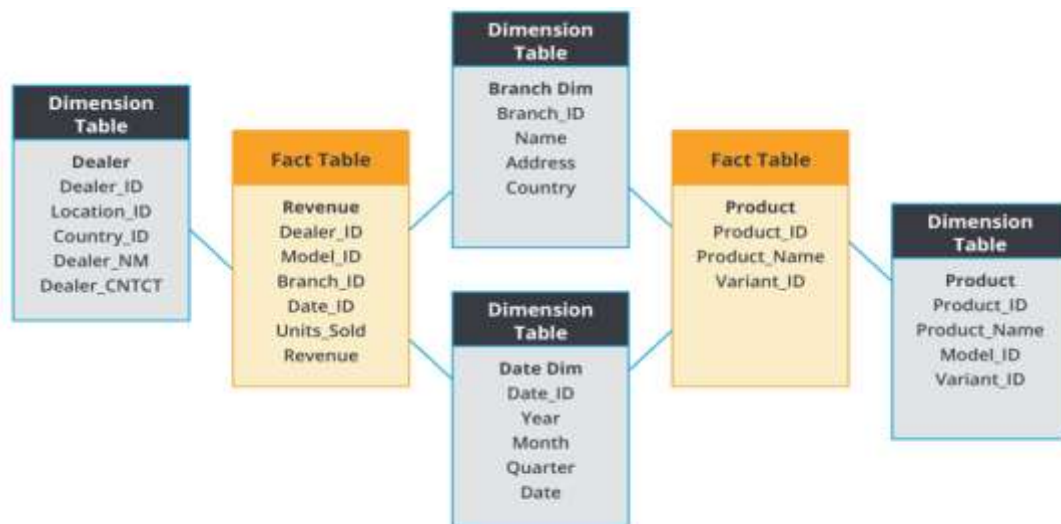
Snowflake schema:

- A snowflake schema is a normalized data model where dimension tables are split into multiple related tables, creating a hierarchical structure.
- While it is more complex than a star schema.



Galaxy schema:

- A Galaxy Schema, also known as a Fact Constellation Schema, is a data warehouse schema design that integrates multiple fact tables sharing common dimension tables.
- It is essentially a collection of interconnected star schemas, making it suitable for handling complex analytical queries across multiple business processes.



Example of Galaxy Schema